

Short-Answer Practice Exam #1

GroupSolve

September 5th, 2026

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- Practice exam #1 includes a speed exam, an accuracy exam, and further problems to pique your interests. This is because I believe in order to excel in math competitions, you need to be fast to come up with difficult ideas.
- The speed exam consists of 20 questions. The suggested time is 40 minutes.
- The accuracy exam consists of 10 questions. The suggested time is 60 minutes.
- You may submit in the google form provided. Solutions will be released at the end of the 2-week period.

1 Speed Round

20 problems in 40 minutes. No extra resources may be provided.

1. If $2m = 200cm$ and $m \neq 0$, find c .
2. Ben was trying to solve for x in the equation $6 + x = 1$. Unfortunately, he was reading upside-down and misread the equation as $1 = x + 9$. What is the positive difference between Ben's answer and the correct answer?
3. Find the number of zeroes at the end of 20^{26} .
4. Anjali and Meili each have a chocolate bar shaped like a rectangular box. Meili's bar is four times as long as Anjali's, while Anjali's is three times as wide and twice as thick as Meili's. What is the ratio of the volume of Anjali's chocolate to the volume of Meili's chocolate?
5. In a school, 60% of the students are boys and 40% are girls. Given that 40% of the boys like math and 50% of the people who like math are girls, what percentage of girls like math?
6. Positive two-digit number $\overline{ab}$ has 8 divisors. Find the number of divisors of the four-digit number $\overline{abab}$.
7. One day, Isaac begins sailing from Marseille to New York City. On the exact same day, Evan begins sailing from New York City to Marseille along the exact same route as Isaac. If Marseille and New York are exactly 3000 miles apart, and Evan sails exactly 40 miles per day, how many miles must Isaac sail each day to meet Evan's ship in 30 days?
8. Kristy is thinking of a number $n < 10^4$ and she says that 143 is one of its divisors. What is the smallest number greater than 143 that could divide n ?
9. If P is a quadratic polynomial with leading coefficient 1 such that $P(1) = 1$, $P(2) = 2$, what is $P(10)$?
10. There are 10 (not necessarily distinct) positive integers with arithmetic mean 10. Determine the maximum possible range of the integers. (The range is defined to be the nonnegative difference between the largest and smallest number within a list of numbers.)
11. A positive integer is called an imposter if it can be expressed in the form $2^a + 2^b$ where a, b are non-negative integers and $a \neq b$. How many almost positive integers less than 2017 are imposters?
12. Alice, Bob, and four other people sit themselves around a circular table. What is the probability that Alice does not sit to the left or right of Bob?
13. How many nonnegative integer pairs (a, b) are there that satisfy $ab = 90 - a - b$?
14. Let $f(x) = |x - 8|$. Let p be the sum of all the values of x such that $f(f(f(x))) = 2$ and q be the minimum solution to $f(f(f(x))) = 2$. Compute $p \cdot q$.
15. How many ways are there to put two identical red marbles, two identical green marbles, and two identical blue marbles in a row such that no red marble is next to a green marble?
16. The tiny island nation of Konistan is a cone with height twelve meters and base radius nine meters, with the base of the cone at sea level. If the sea level rises four meters, what is the surface area of Konistan that is still above water, in square meters?
17. Let $\triangle ABC$ be a triangle. Let M be the midpoint of AB and let P be the projection of A onto BC . If $AB = 20$, and $BC = MC = 17$, compute BP .

18. A gorgeous sequence is a sequence of 1's and 0's such that there are no consecutive 1's. For instance, the set of all gorgeous sequences of length 3 is $\{[1, 0, 0], [1, 0, 1], [0, 1, 0], [0, 0, 1], [0, 0, 0]\}$. Determine the number of gorgeous sequences of length 7.
19. In a right-angled trapezoid $ABCD$, $\angle B = \angle C = 90^\circ$, $AB = 20$, $CD = 17$, and $BC = 37$. A line perpendicular to DA intersects segment BC and DA at P and Q respectively and separates the trapezoid into two quadrilaterals with equal area. Determine the length of BP .
20. A sequence of integers a_i is defined by $a_1 = 1$ and $a_{i+1} = 3i - 2a_i$ for all integers $i \geq 1$. Given that $a_{15} = 5476$, compute the sum $a_1 + a_2 + a_3 + \dots + a_{15}$.

2 Accuracy Round

10 problems in 60 minutes. No extra resources may be provided.

1. A right triangle has two sides of lengths 3 and 4. Find the smallest possible length of the third side.
2. A perfect square, a perfect cube, and a prime number all greater than one have a total sum of 25. What is their product?
3. Mudkip read T pages of a book on Monday and the next T pages of the book on Tuesday. The page numbers read on Monday sum to 810, and the page numbers read on Tuesday sum to 2106. What is the value of T ?
4. According to the Egyptian Metropolitan Culinary Community, food service is delayed on $\frac{2}{3}$ of flights departing from Cairo airport. On average, if flights with delayed food service have twice as many passengers per flight as those without, what is the probability that a passenger departing from Cairo airport experiences delayed food service?
5. Let a , b , and c be positive integers such that $ab + c = 335$ and $bc + a = 316$. Find the sum of all possible values of $a + b + c$.
6. Let N be the smallest positive integer such that all of $\sqrt[2]{2N}$, $\sqrt[3]{3N}$, and $\sqrt[5]{5N}$ are positive integers. Find the smallest positive integer T such $\sqrt[7]{TN}$ is also a positive integer.
7. Let $ABCD$ be a parallelogram with area 140. Points M and N lie on $\overline{AB}$ and $\overline{CD}$ such that $AMCN$ is a rhombus. Given that $MN = 10$ and $AC = 20$, find BD^2 .
8. Find the number of 9-digit positive integers such that each pair of adjacent digits form a multiple of 7.
9. Let $P(x)$ and $Q(x)$ be monic cubic polynomials such that

$$\begin{cases} P(x) = 3 \cdot Q(x) & \text{for } x = 1, 2, 3, \text{ and} \\ Q(x) = 3 \cdot P(x) & \text{for } x = 7, 8, 9. \end{cases}$$

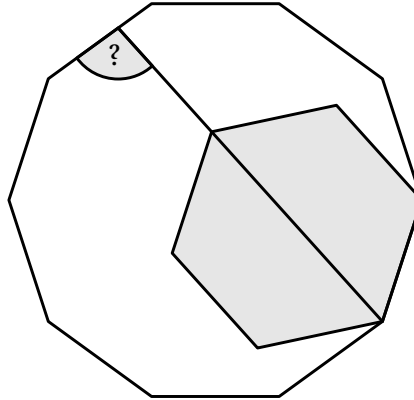
Compute $Q(5)$.

10. For all real numbers x , define $f(x) = x^2 + 60x + k$ for some real constant k . Find the maximum possible value of k such that $f(f(x))$ has at least three distinct real roots.

3 Extra Problems

5 problems in ∞ minutes. No extra resources may be provided.

1. Two overlapping regular polygons are shown below. What is the marked angle, in degrees?



2. For real numbers a, b , and c , suppose that $\frac{a+b}{c}$ and $\frac{b+c}{a}$ are the roots of the polynomial $x^2 - 574x + 17$. Find $\frac{a+c}{b}$.
3. In the coordinate plane, Marcus travels from $(0,0)$ to $(7,2)$ by moving 1 unit in the positive x or y direction each second, while Jadon travels from $(0,2)$ to $(7,0)$ by moving 1 unit in the positive x or negative y direction each second. If they begin their journeys at the same time, in how many ways can Marcus and Jadon choose their paths so that they will meet at some lattice point before reaching their destinations?
4. Compute the number of ordered triples of positive integers (a, b, c) such that
- a divides bc ,
 - b divides ca ,
 - c divides ab , and
 - $abc = 2^{12} \cdot 3^{13}$.
5. The *unit fraction decomposition* of a positive rational $a < 1$ is the unique list $[a_1, a_2, \dots, a_n]$ with each a_i being the smallest positive integer satisfying

$$\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_{i-1}} + \frac{1}{a_i} \leq a.$$

What is the sum of all positive integers N for which there are exactly 18 unit fraction decompositions of the form $[a_1, a_2, \dots, N]$?